Exploring Dark Energy in Conformal Cyclic Cosmology

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Abstract

We explore the role of dark energy within the framework of Conformal Cyclic Cosmology. We first investigate how conformal rescaling relating successive aeons may modify the effective cosmological constant through contributions inherited from the previous aeon together with those induced by the conformal rescaling itself. Furthermore, we discuss how non-Markovian correlations between successive aeons could allow the effective cosmological constant to evolve dynamically across cosmic cycles, offering a possible perspective on the cosmological constant problem in which the apparent fine-tuning of its observed small value may emerge through cumulative evolution across successive aeons. We then consider the implications of these conformal contributions for Early Dark Energy, finding that consistency excludes a non-phantom asymptotic equation of state for the preceding aeon. We further argue that recent Big Bang Nucleosynthesis constraints apply to the effective equation of state parameter rather than directly to the underlying quintessence field. This naturally introduces a parameter degeneracy that may allow the scalar field to remain in the slow-roll regime during Big Bang Nucleosynthesis, while remaining compatible with current observations. The overall aim is to promote discussions regarding the dark energy sector of Conformal Cyclic Cosmology.

1 Introduction

The observed accelerated expansion of the universe is commonly attributed to a small but nonzero cosmological constant, Λ [1]. Despite the remarkable success of this simple description, the cosmological constant gives rise to one of the deepest puzzles in modern physics [1–8]. In quantum field theory, the zero-point fluctuations of quantum fields contribute to the vacuum energy, which gravitates as an effective cosmological constant. However, straightforward estimates of this vacuum energy exceed the observed value by many orders of magnitude, requiring an extraordinary degree of fine-tuning between the bare cosmological constant and quantum corrections to reproduce the tiny value inferred from cosmological observations. This severe discrepancy constitutes the well-known old cosmological constant problem [2, 5]. More generally, despite the observational success of the ΛCDM model, the fundamental origin and nature of dark energy remains unknown. This naturally motivates exploring whether alternative cosmological frameworks can provide new perspectives on the same.

Cyclic cosmologies provide one such framework by extending cosmological evolution across multiple cosmic epochs [9–12]. In particular, Conformal Cyclic Cosmology (CCC) [13–16] proposes that the universe evolves through an infinite sequence of successive aeons, where the asymptotic future of one aeon is conformally related to the Big Bang of the next. In the orthodox formulation of CCC, the transition between successive aeons is governed by a conformal rescaling of the metric, implying that the dynamics of each aeon depend only on the immediately preceding one and are therefore effectively Markovian. Since this conformal crossover relates the geometry and energy content of successive aeons, it is natural to ask whether it also

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modifies the effective dark-energy sector of the newly emerging universe. In particular, the conformal transformation may introduce additional contributions to the effective energy-momentum tensor, thereby altering both the cosmological-constant description of dark energy and its dynamical realizations.

Motivated by these considerations, we first investigate whether the conformal rescaling relating successive aeons modifies the effective cosmological constant of the newly emerging universe. We argue that it receives additional contributions arising both from the vacuum-energy content inherited from the previous aeon and from the conformal rescaling itself. Furthermore, if the conformal factor possesses non-Markovian correlations across successive aeons [17, 18], these contributions could evolve from one cycle to the next, providing an exploratory perspective on the cosmological constant problem in which the observed small value of Λ may emerge through cumulative evolution across successive aeons rather than being fixed once and for all.

We then turn to dynamical dark-energy models, focusing on Early Dark Energy (EDE)[1], which is conventionally modeled by a quintessence field [21–23]. We investigate how the inherited dark-energy component from the previous aeon and the conformally induced dark-energy contribution modify the effective EDE sector. In particular, we find that consistency of the inherited dark-energy sector excludes a non-phantom asymptotic equation of state for the preceding aeon. Furthermore, the Big Bang Nucleosynthesis [24–26](BBN) constraints derived in [27], which determine the phenomenologically preferred range of the EDE equation of state parameter (EOS), now apply to the effective EOS in our framework rather than directly to the underlying quintessence field, leading to a natural parameter degeneracy. This allows the underlying quintessence field to remain in the slow roll regime during BBN, with an EOS of -1, while simultaneously satisfying phenomenologically preferred values.

The remainder of this manuscript is organized as follows. In Sec.2, we briefly review the CCC framework. A discussion of the conserved quantities associated with the crossover surface, together with a detailed derivation, is provided in Appendix A. In Sec.3, we investigate the implications of conformal rescaling for a cosmological-constant description of dark energy, arguing that the conformal transformation naturally induces additional contributions to the effective cosmological constant and discussing how a possible history dependence[2] of the conformal factor could provide an exploratory perspective on the cosmological constant problem. In Sec.4, we turn to dynamical dark-energy models, focusing on EDE, and examine how the conformal contributions modify the effective EOS and its interpretation in light of recent BBN constraints. Finally, in Sec.5, we summarize and discuss the present work.

2 Conformal Cyclic Cosmology

The key idea in CCC [13–16] is that, in the remote future of an aeon [3], the universe effectively "forgets" its overall scale. This allows the asymptotic future of one aeon to be conformally matched to the Big Bang of the next, despite the infinite expansion involved [28]. Such a matching is possible because a large class of space-times admit conformal compactifications in which both the initial singularity and the future infinity can be represented as smooth spacelike hypersurfaces.

For this conformal identification to be physically meaningful, it is essential that no massive particles persist into the asymptotic future. Massive particles introduce an intrinsic scale and would therefore obstruct the conformal invariance required for the matching. In contrast, a universe populated only by massless degrees of freedom (effectively conformally invariant) cannot distinguish overall scale, making the transition between aeons well-defined.

[1] For a recent review regarding EDE, see [19, 20].

[2] In orthodox CCC, the conformal factor is determined solely by the geometry and physical content of the immediately preceding aeon. By history dependence, we instead mean that the conformal factor depends on the cumulative geometry, dynamics, and physical content of all previous aeons, as a consequence of non-Markovian correlations across them.

[3] Throughout this work, we use the term aeon to denote a single cycle in a CCC universe.

Now, in more detail concerning CCC [13–16] [4],in which the relation between successive aeons is established through a conformal rescaling that identifies the future null infinity of one aeon with the Big Bang hypersurface of the next:

$$g_{\mu\nu}^{(n+1)} = \left(\Omega^{(n)}\right)^{-4} g_{\mu\nu}^{(n)}. \tag{1}$$

Moreover, one can introduce a conformally invariant "bandage" region where the metric of each aeon can be related to a bridging metric $g_{ab}^{\tilde{n}}$ via

$$g_{ab}^{n} = \left(\Omega^{(n)}\right)^{2} g_{ab}^{\tilde{n}}, \quad g_{ab}^{(n+1)} = \left(\Omega^{(n)}\right)^{-2} g_{ab}^{\tilde{n}}. \tag{2}$$

Now, the conformal rescaling of the previous aeon's scalar curvature R to the scalar curvature R of the bridging metric is

$$\left(\Box + \frac{1}{6}R^{(\tilde{n})}\right)\Omega^{(n)} = \frac{1}{6}R^{(n)}\left(\Omega^{(n)}\right)^{3}. \tag{3}$$

Penrose proposes that this must be satisfied with the restriction $R^{(n)} = 4\lambda^{n} = R^{(\tilde{n})}$, where $\lambda^{(n)}$ is the cosmological constant. This then yields a second order hyperbolic equation determined solely by quantities in the past aeon, called the Phantom Field Equation [14], which fixes the conformal factor up to two pieces of initial data on $\mathcal{C}$. Nemann [30], Tod [31], Nurowski [32] all differ from Penrose by not enforcing $R^{\tilde{n}} = R^{n}$, but rather that R is possibly some other constant.

The Einstein equation of the previous aeon in the bandage region is $G_{ab}^{(n)} = \kappa T_{ab}^{(n)}$, where the energy-momentum $T_{ab}^{(n)}$ must be divergence-free, trace-free (zero rest-mass) and can be taken to be a conformal density of weight -2 to preserve the divergence-free condition throughout $\mathcal{C}$. Then this propagates through $\mathcal{C}$ to yield

$$T_{ab}^{(n+1)} = \left(\Omega^{(n)}\right)^{4} T_{ab}^{(n)}. \tag{4}$$

Then the Einstein equation in the future aeon is

$$\begin{aligned} G_{\mu\nu}^{n+1} - \lambda^{(n+1)} g_{\mu\nu}^{(n+1)} = \frac{8\pi G}{c^{4}}\left[-\Omega^{-4}T_{\mu\nu}^{n} + 4\frac{\nabla_{\mu}\nabla_{\nu}\Omega}{\Omega} + \frac{\nabla_{\mu}\Omega\nabla_{\nu}\Omega}{\Omega^{2}} + \left(8\frac{\nabla_{\sigma}\Omega\nabla^{\sigma}\Omega}{\Omega^{2}} - \frac{4\Box\Omega}{\Omega}\right)g_{\mu\nu}\right] \\ = -\frac{8\pi G}{c^{4}}U_{\mu\nu}^{(n+1)}. \end{aligned} \tag{5}$$

where $U_{ab}^{(n+1)}$ is the total energy-momentum of the future aeon. It is clear that this has a trace term in general and thus matter can appear in the future aeon. Penrose postulates this corresponds to dark matter [14]. Furthermore, he decomposes $U_{ab}^{(n)}$ into three distinct parts

$$U_{ab}^{(n+1)} = T_{ab}^{(n+1)} + T_{ab}^{(n+1)}[\Omega^{(n+1)}] + W_{ab}^{(n+1)}. \tag{6}$$

Here $T_{ab}^{(n+1)}$ is the contribution coming from $T_{ab}^{(n)}$, the energy momentum content of the previous aeon. $T_{ab}^{(n+1)}[\Omega^{(n)}]$ is the contribution coming from the scalar field $\Omega^{(n)}$ that is also called the new-improved energy momentum tensor and is trace-free. $W_{ab}^{(n)}$ is the portion that contains the trace-term and the symmetric part is taken so that the Einstein equations in the future aeon are satisfied.

Furthermore, it can be shown (see Appendix A) that the total energy of the preceding aeon is conserved across the crossover surface [33]. A particularly important contribution to this conserved energy arises from

[4]It should be noted that orthodox CCC does not invoke an inflationary phase. Within CCC, there is no horizon or flatness problem in the conventional sense, and the observed near scale-invariance of the cosmic microwave background (CMB) temperature anisotropies is instead attributed to processes occurring in the preceding aeon, rather than to an early phase of exponential expansion.

[5]In Penrose's original formulation [12], CCC involves information loss within black holes, a feature that remains controversial. However, alternative formulations have been proposed that aim to preserve unitarity,for example, the work [29] explores a version of CCC compatible with quantum information conservation.

the Hawking radiation emitted by black holes formed within that aeon.

This energy is predicted to leave detectable anomalies in the CMB of our present aeon. In particular, this will induce distinct regions in the CMB with higher temperatures by order of magnitude 10^{-3} ($\delta T/T \sim 10^{-3}$). Penrose calls such spots "Hawking spots" [33] and proposes that they arise due to the energy carried by the Hawking radiation of the previous aeon.

Let us now try to theoretically estimate $\delta T/T$ of Hawking Spots. Note that masses of the largest galaxy clusters correspond to

$$M_{gc} \sim 10^{15} M_{\odot}. \tag{7}$$

To calculate the temperature elevation on the CMB map on the Hawking spots we start from the density at the time of last scattering

$$\rho_{LS} \sim \frac{\rho_0}{a_{LS}^3} \sim 1.2 \cdot 10^{-20} \mathrm{kg/m^3}. \tag{8}$$

The maximal average density of the injected mass (energy) from the evaporation of the largest galaxy clusters in the previous aeon is given by

$$\delta\rho \sim \frac{3M_{gc}}{4\pi(\eta_{LS}^c)^3 c^3 a_{LS}^3} \sim 10^{-22} \mathrm{kg/m^3}. \tag{9}$$

We now have

$$\frac{\delta T}{T} \sim \frac{\delta\rho}{4\rho_{LS}} \sim 10^{-3}, \tag{10}$$

exactly corresponding to the observed CMB anomalies.

3 Cosmological Constant and Conformal Cyclic Cosmology

A distinctive feature of CCC [13–16] is that the conformal crossover relating successive aeons modifies the effective gravitational dynamics of the newly emerging universe. Within the standard CCC framework reviewed in Sec.2, the Einstein field equations naturally acquire additional contributions arising from the conformal transformation. In this work, we interpret these terms as effective contributions to the stress-energy tensor and propose that they may be regarded as part of the dark-energy sector of the newly emerging aeon. This naturally motivates the introduction of an effective cosmological constant, incorporating both the fundamental cosmological constant appearing in the Einstein field equations and additional contributions induced by the conformal transformation.

In particular, from eqn.(5), we observe that the Einstein equations in the new aeon can be written in terms of an effective stress-energy tensor that includes contributions arising from the conformal factor. Explicitly, this takes the form

$$U_{\mu\nu}^{(n+1)} = \Omega^{-4} T_{\mu\nu}^n - 4\frac{\nabla_\mu\nabla_\nu\Omega}{\Omega} - \frac{\nabla_\mu\Omega\nabla_\nu\Omega}{\Omega^2} - \left(8\frac{\nabla_\sigma\Omega\nabla^\sigma\Omega}{\Omega^2} - \frac{4\Box\Omega}{\Omega}\right) g_{\mu\nu}. \tag{11}$$

Among the various conformally induced contributions to the effective stress-energy tensor, we consider only the term proportional to the metric.

$$T_{\mu\nu}^{(\mathrm{DE},\Omega)} = \left(-8\frac{\nabla_c\Omega\nabla^c\Omega}{\Omega^2} + \frac{4\Box\Omega}{\Omega}\right) g_{\mu\nu}. \tag{12}$$

Since this contribution satisfies $\rho_{(\mathrm{DE},\Omega)} = -p_{(\mathrm{DE},\Omega)}$., it naturally admits the interpretation of an effective cosmological-constant component. The remaining terms are not considered further here and may instead describe other components of the effective energy-momentum tensor generated by the conformal crossover.

The corresponding energy density is given by

$$\rho_{(\mathrm{DE},\Omega)} = -8\frac{\nabla_c\Omega\nabla^c\Omega}{\Omega^2} + \frac{4\Box\Omega}{\Omega}. \tag{13}$$

For a homogeneous conformal factor $\Omega = \Omega(t)$, this becomes

$$\rho_{(\mathrm{DE},\Omega)} = 4\frac{\ddot{\Omega} + 3H\dot{\Omega}}{\Omega^2} - 8\frac{\dot{\Omega}^2}{\Omega^2}. \tag{14}$$

Thus, the effective dark energy density is dynamically determined by the evolution of the conformal factor and is sensitive to the expansion history of the universe.

In the present framework, the Einstein equations in the new aeon contain an effective dark energy contribution arising from the conformal factor. As shown above, the term proportional to the metric can be identified with an effective cosmological constant, yielding

$$\Lambda_{\mathrm{eff}}^{(n+1)} = \lambda^n + 8\pi G(\rho_{\mathrm{vac}}^{(n)} + \rho_{\mathrm{DE},\Omega}^{(n)}) \tag{15}$$

where $\rho_{\mathrm{DE}}^{(n)}$ is determined by the conformal factor $\Omega^{(n)}$. Unlike the standard picture, this contribution is dynamical and depends on the evolution of Ω.

A central feature of the present proposal is that the conformal factor is not determined solely by the immediately preceding aeon. Instead, we allow for a history-dependent structure in which

$$\Omega^{(n)} = \mathcal{F}[g^{(1)}, g^{(2)}, ...g^{(n)}] \tag{16}$$

so that the conformal transformation between aeons encodes cumulative information about the geometry of the entire past sequence. In this sense, the evolution across aeons is non-Markovian.

Despite this generalization, the conformal factor is not arbitrary. In each aeon it is constrained to satisfy the conformal field equation inherited from the CCC framework,

$$\left(\Box + \frac{1}{6}R^{(\tilde{n})}\right)\Omega^{(n)} = \frac{1}{6}R^{(n)}\left(\Omega^{(n)}\right)^3. \tag{17}$$

which restricts the admissible class of functions $\Omega^{(n)}$. Thus, the space of possible conformal factors across aeons is not unrestricted, but consists of solutions to a common geometric equation, differing only in their initial data and history dependence.

Within this allowed solution space, we propose that the sequence $\{\Omega^{(n)}\}$ evolves iteratively across aeons in such a way that the effective cosmological constant is progressively reduced. Schematically, this may be expressed as an update rule of the form

$$\Omega^{(n+1)} = \Omega^{(n)} + \Delta G(n) \tag{18}$$

where the increment $G(n)$ depends on the geometric and dynamical properties of the previous aeons. In particular, the evolution is such that it drives the system toward configurations that reduce the magnitude of Λ_{eff}.

This induces a corresponding evolution of the effective cosmological constant,

$$\Lambda_{\mathrm{eff}}^{(0)} \to \Lambda_{\mathrm{eff}}^{(1)} \to \cdots \to \Lambda_{\mathrm{eff}}^{(n)}, \tag{19}$$

which need not be constant across aeons. Instead, the sequence can exhibit a relaxation behavior toward a stable regime.

The key idea is that, within the space of admissible solutions to the conformal field equation, there exist configurations of Ω that yield significantly smaller values of Λ_{eff}. The non-Markovian dependence allows the system to effectively "explore" this space across successive aeons. As a result, after sufficiently many cycles, the evolution selects an effective conformal factor $\Omega^{(n)}$ that corresponds to a near-optimal configuration in which the cosmological constant is dynamically suppressed.

In this picture, the observed small value of the cosmological constant does not arise from a finely tuned cancellation at a single epoch, but rather emerges as the outcome of a cumulative selection process across aeons. The conformal factor acts as a carrier of non-Markovian correlations across aeons, and its constrained evolution under the field equation leads naturally to a form of dynamical fine-tuning.

To make this idea more concrete, suppose that the existence of a universe capable of supporting complex structure and intelligent observers requires the cosmological constant to take a value satisfying some special condition. In general, such a condition may be represented abstractly by a complex and non-trivial equation of the form

$$F(\Lambda, \alpha_1, \alpha_2, \dots) = 0, \tag{20}$$

where α_i denote other fundamental parameters of the theory. The observed value of Λ in this aeon would then correspond to one particular root of this equation. In the present framework, there is no need to assume that the cosmological constant is fixed to this special value from the outset. Rather, the history-dependent conformal factor acts as a memory-carrying degree of freedom that dynamically adjusts the effective cosmological constant from one aeon to the next. Across sufficiently many cycles, the accumulated non-Markovian evolution drives Λ_{eff} toward a stable root of F, namely one lying within the narrow range compatible with long-lived stars, galaxy formation, and ultimately the emergence of intelligent life. In this sense, what appears as fine-tuning within a single aeon is reinterpreted as a cosmological optimization process operating over an arbitrarily long sequence of aeons.

As an illustrative example, we consider a simple iterative evolution of the conformal contribution across aeons of the form

$$\rho_\Omega^{(n+1)} = \rho_\Omega^{(n)} - \alpha \Lambda_{\text{eff}}^{(n)}, \tag{21}$$

where $\alpha > 0$ controls the rate of adjustment. This relation captures the idea that the conformal sector responds to the residual effective cosmological constant in each aeon, reducing its magnitude in subsequent cycles.

Substituting $\Lambda_{\text{eff}}^{(n)} = \lambda^{(n)} + 8\pi G(\rho_{\text{vac}}^{(n)} + \rho_\Omega^{(n)})$,one finds that this leads to a relaxation behavior in which the effective cosmological constant is progressively suppressed across successive aeons, with the residual value decreasing approximately exponentially with the number of cycles.

We stress, however, that this represents only the most trivial example of such an iterative mechanism. In particular, this simple linear model generically drives the system toward $\Lambda_{\text{eff}} \to 0$, and therefore does not by itself account for the observed small but nonzero value of the cosmological constant. For this reason, it should not be regarded as a realistic description of the underlying dynamics.

Rather, the purpose of this example is to illustrate the general principle that a history-dependent conformal factor can induce a cumulative relaxation of vacuum energy across aeons. In a more complete treatment, the evolution of ρ_Ω would be governed by the conformal field equation together with its non-Markovian dependence on the full geometric history, and is expected to exhibit significantly more intricate and nontrivial behavior. Constructing such a model is left for future work.

The key point is that, even without specifying the detailed dynamics, this framework provides a natural mechanism by which large vacuum energy contributions may be progressively suppressed over successive cycles. The conformal factor, constrained by its field equation yet sensitive to past evolution, acts as a memory-carrying degree of freedom that drives the system toward regimes of small effective cosmological constant.

Now, the gravitational sector plays a central role in CCC, as it is one of the few physical sectors expected to survive between the conformal crossover between successive aeons. Unlike massive particles, which ultimately become irrelevant in the asymptotic future (according to the assumptions of orthodox CCC), gravitons are massless and therefore remain compatible with the conformal symmetry that underpins the transition from

one aeon to the next. Gravitational radiation generated throughout an aeon, including primordial tensor perturbations and waves produced by compact astrophysical sources, thus represents a potential carrier of information cross the conformal boundary. In the present framework, an additional physical significance of the gravitational wave sector lies in the information they encode, which ultimately influences the non-Markovian correlations between successive aeons. In particular, including the influence of the gravitational sector now yields

$$\Omega^{(n)} = \mathcal{F}[g^{(1)}, g^{(2)}, ...g^{(n)}, \Psi^{(1)}_{\rm GW}, \Psi^{(2)}_{\rm GW}, \dots \Psi^{(n)}_{\rm GW}], \tag{22}$$

where $\Psi^{(k)}_{\rm GW}$ denotes the state of the gravitational sector of the kth, as it approaches the crossover surface. This generalization explicitly incorporates the cumulative influence of gravitational radiation into the inter-aeonic memory, allowing the gravitational sector to modify the non-Markovian evolution across successive aeons.

Now, in contrast to the standard inflationary scenario, where the tensor spectrum is determined solely by the dynamics of the inflating spacetime, the gravitational sector in our framework carries an additional memory of the conformal crossover. As a consequence, the primordial tensor power spectrum is expected to acquire a scale-dependent correction, which may be parametrized as

$$\mathcal{P}_T(k) \to \mathcal{P}_T(k)[1 + \delta_{\rm mem}(k)], \tag{23}$$

where $\delta_{\rm mem}(k)$ represents the scale-dependent contribution arising from the non-Markovian effects. In general, this correction encapsulates the cumulative influence of the inter-aeonic correlations on tensor modes, with its precise form depending on the microscopic dynamics responsible for the transfer of information across the conformal crossover. Since the tensor-to-scalar ratio is directly determined by the ratio of the primordial tensor spectrum relative to the scalar spectrum, any modification of $\mathcal{P}_T(k)$ directly affects r. Assuming that the scalar power spectrum remains unaffected, the tensor-to-scalar ratio is correspondingly modified according to

$$r \to r[1 + \delta_{\rm mem}(k_0)], \tag{24}$$

where k_0 denotes the CMB pivot scale at which the tensor-to-scalar ratio is conventionally quoted. Consequently, the non-Markovian corrections proposed in this framework provide a direct observational channel through which inter-aeonic memory effects could be constrained. Any statistically significant deviation of the measured tensor-to-scalar ratio from the standard inflationary prediction, particularly if accompanied by a characteristic scale dependence in the tensor spectrum, could therefore serve as evidence for such inter-aeonic correlations. While the explicit functional form of $\delta_{\rm mem}(k)$ depends on the exact nature of the non-Markovian correlation, which is beyond the scope of the present work, the above parametrization provides a model-independent framework for investigating the consequences of these effects.

A further appealing feature of the present non-Markovian extension is that it does not significantly alter the observational predictions of standard CCC, in particular the temperature profile of Hawking spots.

In the Markovian setting, the conserved quantity associated with the twistor construction yields

$$M = 4\pi G m \tag{25}$$

which underlies the estimate of the temperature anisotropy

$$\frac{\delta T}{T} \sim 10^{-3} \tag{26}$$

In the presence of history dependence, the conformal structure is modified and, correspondingly, the conservation law receives corrections. At an effective level, this may be expressed as

$$M = 4\pi G m + K \tag{27}$$

where K encodes contributions arising from the non-Markovian dependence of the conformal factor on the past history of the geometry (or, in the case of isolated sources, the history of the mass distribution). By

construction, $K \to 0$ in the Markovian limit.

For modest non-Marknovian correlations, $K << m \sim 10^{15} M_\odot$ (where $M_\odot$ denotes the solar mass and $10^{15} M_\odot$ is the estimate for the largest galaxy clusters),the history-dependent contribution does not modify the order of magnitude of the predicted temperature anisotropy. Consequently, the leading-order prediction $\delta T/T \sim 10^{-3}$ remains essentially unchanged. This demonstrates that the inclusion of history dependence preserves the principal observational predictions of CCC while introducing only subleading corrections.

In summary, the introduction of a history-dependent conformal factor leads to a natural modification of the effective stress-energy tensor, giving rise to a dynamically generated dark energy component whose magnitude evolves across aeons. While the precise evolution of this contribution is governed by the conformal field equation and its non-Markovian dependence on past geometry, even simple illustrative models demonstrate the possibility of cumulative suppression of the effective cosmological constant without requiring fine-tuned cancellations at a single epoch. Importantly, this framework preserves the key observational features of standard CCC, corrections to quantities such as the Hawking spot temperature anisotropy [33] are found to be extremely suppressed, leaving leading-order predictions unchanged. Thus, the non-Markovian extension provides a consistent and observationally viable generalization of CCC, in which the fine-tuning of the cosmological constant can be understood as the result of long-term dynamical evolution across aeons.

4 Early Dark Energy in Conformal Cyclic Cosmology

Having discussed the implications of conformal rescaling for a cosmological-constant description of dark energy, we now turn to dynamical dark-energy models. EDE [19, 20] provides a natural framework in which to investigate the consequences of the additional conformal contributions arising within the CCC framework. As we shall show, these contributions alter the effective EDE sector, modifying its EOS.

Conventionally, EDE [19, 20] is modeled by a canonical quintessence scalar field ϕ [21–23]. The energy density and pressure of the field are given by

$$\rho_{\rm Q} = \frac{\dot{\phi}^2}{2} + V(\phi) \tag{28}$$

and

$$p_{\rm Q} = \frac{\dot{\phi}^2}{2} - V(\phi), \tag{29}$$

respectively, yielding the equation of state

$$\omega_{\rm Q} = \frac{\frac{\dot{\phi}^2}{2} - V(\phi)}{\frac{\dot{\phi}^2}{2} + V(\phi)}. \tag{30}$$

Hereafter, the subscript Q denotes quantities associated exclusively with the quintessence-field contribution.

The dynamics of the scalar field are governed by the Klein-Gordon equation in an expanding FLRW universe,

$$\ddot{\phi} + 3H\dot{\phi} + V_{,\phi} = 0, \tag{31}$$

where the Hubble friction term $3H\dot{\phi}$, suppresses the field evolution at early times. Consequently, for redshifts $z \gtrsim z_c$, the scalar field remains Hubble frozen, with its kinetic energy much smaller than its potential energy, $\dot{\phi}^2 << V(\phi)$. The EDE component therefore behaves approximately as a cosmological constant, giving an equation of state $\omega_{\rm Q} \simeq -1$. As the universe expands, the Hubble friction weakens and the field eventually thaws around the critical redshift z_c, after which it begins to roll down its potential. The kinetic energy then becomes increasingly important, causing the equation of state to deviate from -1 and marking the onset of the characteristic transient evolution of the EDE component.

Early Dark Energy models were originally proposed as a mechanism to alleviate the Hubble tension by introducing a transient dark-energy component that contributes a small fraction of the total energy density prior to recombination, thereby reducing the sound horizon while leaving the late-time expansion history largely unaffected. To preserve the successful predictions of standard cosmology, the onset of the EDE transition is typically chosen to occur at a critical redshift z_c well after the epoch of BBN, with $z_c << z_{\rm BBN}$. Consequently, conventional EDE models predict that the scalar field remains Hubble frozen throughout the BBN epoch, implying an equation of state very close to that of a cosmological constant, $\omega_{\rm Q} \simeq -1$. However, recent BBN analysis [27] based on a phenomenological constant-ω EDE find that the EOS during the BBN epoch is constrained to the 68% confidence interval

$$\omega_{\rm EDE} \simeq [-0.918, -0.432]. \tag{32}$$

Although this parametrization does not fully describe the dynamics of thawing scalar-field EDE models, it nevertheless provides an informative benchmark for the EOS allowed during BBN. Furthermore, it indicates a substantial departure from the slow-roll value already during the BBN era. This presents a potential challenge for conventional quintessence-based EDE models, in which the scalar field is expected to remain Hubble-frozen until well after BBN. The apparent tension motivates reconsidering the effective dark-energy sector within frameworks that extend beyond the conventional quintessence description.

Within the CCC framework, the effective Early Dark Energy component is not described solely by the quintessence field of the new aeon. Instead, it also receives contributions from the dark-energy density inherited from the previous aeon together with additional stress-energy generated by the conformal transformation relating successive aeons. Consequently, the equation of state constrained by BBN corresponds to that of the effective EDE sector rather than that of the quintessence field alone.

Concretely, let us denote the EOS at the onset of the classical emergence of an aeon by $\omega_0^{(n+1)}$. For simplicity, we assume that the asymptotic EOS of the $n^{\rm th}$ aeon, ω_∞ is conformally invariant i.e, remains unchanged under the conformal rescaling. Since the energy density in the $n^{\rm th}$ aeon scales as

$$\rho^n \propto a^{-3(1+\omega_\infty^{(\rm n)})}, \tag{33}$$

the conformal transformation induces an additional factor of $\Omega^{6(1+\omega_\infty)}$ in $\rho^{(\rm n)}$.

Unlike the analysis of the effective cosmological constant presented in Sec.3, where only the metric-proportional contribution in the stress energy tenso was retained, in the present analysis we also include the remaining conformally induced contribution,

$$\Theta_{\mu\nu}^{(\rm DE,\Omega)} = -4\frac{\nabla_\mu\nabla_\nu\Omega}{\Omega} - \frac{\nabla_\mu\Omega\nabla_\nu\Omega}{\Omega^2} - \left(8\frac{\nabla_\sigma\Omega\nabla^\sigma\Omega}{\Omega^2} - \frac{4\Box\Omega}{\Omega}\right) g_{\mu\nu}. \tag{34}$$

which is not proportional to the metric tensor. Consequently, this component does not, in general, satisfy the vacuum equation of state $\omega_{(\rm DE,\Omega)} = -1$. Instead, we interpret $\Theta_{\mu\nu}^{(\rm DE,\Omega)}$ as an additional dynamical dark-energy contribution generated by the conformal crossover. The effective Early Dark Energy sector therefore consists not only of the inherited dark-energy component and the newly generated quintessence field, but also of this conformally induced dynamical contribution.

$$\omega_0^{(n+1)} = \frac{\Omega^{6(1+\omega_\infty^{(\mathrm{n})})}\omega_\infty^{(\mathrm{n})}\rho^{(n)} + \omega_{\mathrm{Q}}\rho_{\mathrm{Q}}^{(n+1)} + p_{(\mathrm{DE},\Omega)}^{(n)}}{\Omega^{6(1+\omega_\infty^{(\mathrm{n})})}\rho^{(n)} + \rho_{\mathrm{Q}}^{(n+1)} + \rho_{\Omega}^{(n)}}$$

$$= \frac{\Omega^{6(1+\omega_\infty^{(\mathrm{n})})}\omega_\infty^{(\mathrm{n})}\rho^{(n)} + \omega_{\mathrm{Q}}\rho_{\mathrm{Q}}^{(n+1)}}{\Omega^{6(1+\omega_\infty^{(\mathrm{n})})}\rho^{(n)} + \rho_{\mathrm{Q}}^{(n+1)}} \left(\frac{\Omega^{6(1+\omega_\infty^{(\mathrm{n})})}\rho^{(n)} + \rho_{\mathrm{Q}}^{(n+1)}}{\Omega^{6(1+\omega_\infty^{(\mathrm{n})})}\rho^{(n)} + \rho_{\mathrm{Q}}^{(n+1)} + \rho_{(\mathrm{DE},\Omega)}^{(n)}} \right) + \frac{\omega_{(\mathrm{DE},\Omega)}^{(n)}\rho_{(\mathrm{DE},\Omega)}^{(n)}}{\Omega^{6(1+\omega_\infty^{(\mathrm{n})})}\rho^{(n)} + \rho_{\mathrm{Q}}^{(n+1)} + \rho_{(\mathrm{DE},\Omega)}^{(n)}}$$

$$= \omega_\infty^{(\mathrm{n})} \left(\frac{\Omega^{6(1+\omega_\infty^{(\mathrm{n})})}\rho^{(n)}}{\Omega^{6(1+\omega_\infty^{(\mathrm{n})})}\rho^{(n)} + \rho_{\mathrm{Q}}^{(n+1)}} \right) \left(\frac{\Omega^{6(1+\omega_\infty^{(\mathrm{n})})}\rho^{(n)} + \rho_{\mathrm{Q}}^{(n+1)}}{\Omega^{6(1+\omega_\infty^{(\mathrm{n})})}\rho^{(n)} + \rho_{\mathrm{Q}}^{(n+1)} + \rho_{(\mathrm{DE},\Omega)}^{(n)}} \right)$$

$$+ \frac{\rho_{\mathrm{Q}}^{(n+1)}\omega_{\mathrm{Q}}}{\Omega^{6(1+\omega_\infty^{(\mathrm{n})})}\rho^{(n)} + \rho_{\mathrm{Q}}^{(n+1)}} \left(\frac{\Omega^{6(1+\omega_\infty^{(\mathrm{n})})}\rho^{(n)} + \rho_{\mathrm{Q}}^{(n+1)}}{\Omega^{6(1+\omega_\infty^{(\mathrm{n})})}\rho^{(n)} + \rho_{\mathrm{Q}}^{(n+1)} + \rho_{(\mathrm{DE},\Omega)}^{(n)}} \right) + \frac{\omega_{(\mathrm{DE},\Omega)}^{(n)}\rho_{(\mathrm{DE},\Omega)}^{(n)}}{\Omega^{6(1+\omega_\infty^{(\mathrm{n})})}\rho^{(n)} + \rho_{\mathrm{Q}}^{(n+1)} + \rho_{(\mathrm{DE},\Omega)}^{(n)}}. \tag{35}$$

Defining

$$y = \frac{\Omega^{6(1+\omega_\infty^{(\mathrm{n})})}\rho^{(n)} + \rho_{\mathrm{Q}}^{(n+1)}}{\Omega^{6(1+\omega_\infty^{(\mathrm{n})})}\rho^{(n)} + \rho_{\mathrm{Q}}^{(n+1)} + \rho_{(\mathrm{DE},\Omega)}^{(n)}}; \quad x = \frac{\Omega^{6(1+\omega_\infty^{(\mathrm{n})})}\rho^{(n)}}{\Omega^{6(1+\omega_\infty^{(\mathrm{n})})}\rho^{(n)} + \rho_{\mathrm{Q}}^{(n+1)}} \tag{36}$$

one finds

$$\omega_0^{(n+1)} = xy\omega_\infty^{(n)} + (1-x)y\omega_{\mathrm{Q}} + (1-y)\omega_{(\mathrm{DE},\Omega)}^{(n)} = xy\left(\omega_\infty^{(n)} - \omega_{\mathrm{Q}}\right) + y(\omega_{\mathrm{Q}} - \omega_{(\mathrm{DE},\Omega)}^{(n)}) + \omega_{(\mathrm{DE},\Omega)}^{(n)} \tag{37}$$

where $\omega_0^{(n+1)}$ denotes the dark energy EOS at the classical emergence of the $(n+1)^{\mathrm{th}}$ aeon, while $\omega_\infty^{(n)}$ denotes the EOS of the nth aeon immediately prior to tunneling through the crossover surface.

The next, $(n+1)^{\mathrm{th}}$ aeon begins in the limit $\Omega \to \infty$. Consequently, the asymptotic EOS of the nth aeon determines the EOS of the EDE component in the subsequent aeon. Three distinct possibilities arise depending on the asymptotic value ω_∞.

If the nth aeon terminates with a non-phantom EOS, $\omega_\infty > -1$, then

$$\Omega^{6(1+\omega_\infty)} \to \infty, \tag{38}$$

which immediately implies $x = y = 1$. The EOS of the EDE component at the beginning of the next aeon therefore becomes

$$\omega_0^{(n+1)} = \omega_\infty > -1. \tag{39}$$

If, instead, the nth aeon ends in the phantom regime [34–37] [6] with $\omega_\infty < -1$, then

$$\Omega^{6(1+\omega_\infty)} \to 0, \tag{40}$$

and hence, $x = 0$. The EOS in the $(n+1)^{\mathrm{th}}$ aeon is then

$$\omega_0^{(n+1)} = y(\omega_{\mathrm{Q}} - \omega_{(\mathrm{DE},\Omega)}^{(n)}) + \omega_{(\mathrm{DE},\Omega)}^{(n)} \tag{41}$$

Finally, if the nth aeon approaches the limiting case $\omega_\infty = -1$, we obtain

$$\omega_0^{n+1} = -xy\,(1+\omega_{\mathrm{Q}}) + y(\omega_{\mathrm{Q}} - \omega_{(\mathrm{DE},\Omega)}^{(n)}) + \omega_{(\mathrm{DE},\Omega)}^{(n)} \tag{42}$$

[6] A recent work [38] has proposed a new variant of CCC wherein, the future spacetime metric becomes conformally invariant at the Big Rip instead of the remote future. Furthermore, such a modification to CCC removes the need for mass decay from the model and makes it consistent with current physical laws. An exploratory discussion relating phantom-energy cosmology and CCC was also presented in [39], where the authors suggested that quantum smoothing of the Big Rip could facilitate a conformal transition to a subsequent aeon.

These relations may now be confronted with observational constraints. In particular, assuming that $\omega_0^{n+1} > -1$ is still satisfied, we note that recent BBN analyses employing a phenomenological constant-equation-of-state EDE model [27] constrain the effective EDE equation of state during the nucleosynthesis epoch to the 68% confidence interval,

$$\omega_0^{(n+1)} \in [-0.918, -0.432]. \tag{43}$$

Although this parametrization does not capture the full dynamics of scalar-field EDE models, it nevertheless provides a useful benchmark. Indeed, in conventional quintessence-based EDE scenarios the scalar field remains Hubble-frozen throughout the BBN epoch, so that its equation of state varies only negligibly and is well approximated by an approximately constant effective value over the relatively short duration of BBN.

For the non-phantom case, the initial EOS of the new aeon coincides directly with the asymptotic EOS of the previous aeon, as seen in eqn.(39). Consequently,

$$\omega_\infty^{(n)} \in [-0.918, -0.432]. \tag{44}$$

It should be noted, however, that this branch is not physically realized within the present framework. For $\omega_\infty > -1$, the inherited dark energy scales as $\Omega^{6(1+\omega_\infty)}$. Since the new aeon begins in the limit $\Omega \to \infty$, this contribution diverges at the crossover. Consequently, the effective dark-energy density of the newly emerging aeon is dominated by an infinite inherited component, and the resulting cosmology does not admit a finite Early Dark Energy sector compatible with the phenomenology of BBN. In particular, the effective energy density no longer corresponds to the finite EDE component constrained by the BBN analysis of [27]. We therefore conclude that the non-phantom branch, $\omega_\infty > -1$, is excluded within the present framework.

By contrast, if the preceding aeon ends in the phantom regime, the observational constraint becomes

$$y(\omega_{\rm Q} - \omega_{({\rm DE},\Omega)}) + \omega_{({\rm DE},\Omega)} \in [-0.918, -0.432], \tag{45}$$

whereas for the limiting case $\omega_\infty = -1$ one finds

$$-xy\,(1+\omega_{\rm Q}) + y(\omega_{\rm Q} - \omega_{({\rm DE},\Omega)}^{(n)}) + \omega_{({\rm DE},\Omega)}^{(n)} \in [-0.918, -0.432]. \tag{46}$$

Eqns.(45) and (46) demonstrate that the BBN constraints are imposed on specific combinations of the phenomenological parameters rather than on the quintessence EOS alone. Consequently, the observations constrain only the effective EOS of the EDE component, while the underlying quintessence field retains additional freedom through its degeneracy with the parameters $\omega_{({\rm DE},\Omega)}, x, y$. The observationally allowed parameter space is therefore determined by the combined values of $\omega_{({\rm DE},\Omega)}$,$\omega_{\rm Q}$, x, and y, rather than by any one of these quantities individually.

This degeneracy is theoretically advantageous because it preserves the conventional slow-roll evolution of the quintessence field while simultaneously accommodating the BBN constraints on the effective EDE equation of state. In standard EDE scenarios, the scalar field is expected to remain Hubble frozen throughout the BBN epoch and only begins to thaw around the critical redshift $z_c \sim 10^3 - 10^4$, which is precisely the regime required for EDE to alleviate the H_0 tension. By contrast, interpreting the BBN constraint as applying directly to the quintessence field would require an equation of state such as $\omega \sim -0.9$ or -0.8 already during BBN, implying that the field must begin thawing at redshifts many orders of magnitude larger than the conventional EDE thawing redshift.

The present framework therefore possesses an important theoretical advantage. The underlying quintessence field may remain in the slow-roll regime during BBN, with

$$\omega_{\rm Q}^{(n+1)} \simeq -1, \tag{47}$$

while the effective EDE EOS $\omega_0^{(n+1)}$ simultaneously satisfies the phenomenologically preferred interval of eqn.(43). This is possible because the effective EDE sector is not determined by the quintessence field alone. Rather, it also receives contributions from the conformally induced dynamical dark-energy component,

characterized by $\omega_{\mathrm{DE},\Omega}$, together with the inherited dark-energy density from the previous aeon. As encoded by the parameter degeneracies of eqn.(45), suitable choices of $x, y, \omega_{\mathrm{Q}}$ and $\omega^{(n)}_{\mathrm{DE},\Omega}$ allow the effective equation of state to differ substantially from that of the underlying quintessence field. Consequently, satisfying the BBN constraints does not require the quintessence field itself to depart from the slow-roll regime. Instead, the required deviation of the effective EDE equation of state is naturally supplied by the conformally induced dynamical dark-energy contribution, allowing the model to reconcile theoretical expectations for quintessence with current phenomenological constraints.

5 Conclusions

In this work, we have investigated several possible roles that dark energy may play within the framework of CCC [13–16]. We have argued that the conformal structure relating successive aeons gives rise to additional contributions to the effective dark-energy sector, originating both from the inherited vacuum-energy content of the preceding aeon and from the conformal transformation itself. These contributions enrich the standard picture of CCC and suggest that dark energy may provide a valuable probe of the physics governing the conformal crossover between aeons.

Our first principal result concerns the cosmological constant within CCC. We have argued that the conformal crossover naturally gives rise to an effective cosmological constant that receives contributions not only from the vacuum energy of the current aeon, but also from the inherited energy content of the preceding aeon together with terms induced by the conformal rescaling itself. Consequently, the observed cosmological constant need not be attributed solely to the vacuum energy of the present aeon, but may instead contain information inherited from the preceding aeon and from the conformal transition itself. This broadens the standard picture of CCC and suggests that the observed value of the cosmological constant may reflect the cumulative effects of the conformal crossover between successive aeons.

To explore a possible dynamical origin for these additional contributions, we proposed a non-Markovian extension of CCC in which successive aeons are linked through non-Markovian correlations associated with the conformal factor. Within this framework, the effective cosmological constant is no longer determined independently in each aeon, but instead evolves through the cumulative influence of previous cosmic cycles. Such inter-aeonic correlations provide a mechanism by which the effective vacuum energy may be progressively regulated from one aeon to the next, potentially driving it toward the remarkably small value observed today without requiring extreme fine tuning within a single cosmological epoch. Although speculative, this scenario illustrates how non-Markovian correlations can provide a physically motivated mechanism for dynamically regulating the effective cosmological constant while remaining consistent with the philosophy of CCC.

Furthermore, the introduction of non-Markovian dynamics does not significantly alter the leading observational predictions of Hawking spots made by standard CCC. The conserved quantity associated with the crossover surface receives only extremely small corrections, leading to negligible modifications of the predicted Hawking spot temperature anisotropies [33]. Consequently, the characteristic CCC prediction $\delta T/T \sim 10^{-3}$ remains essentially unchanged. The framework therefore preserves the principal observational successes of CCC while simultaneously introducing new structure in the dark sector.

Moreover, since gravitons remain among the few physical degrees of freedom expected to survive the conformal crossover, they naturally act as carriers of inter-aeonic information in the proposed non-Markovian extension of CCC. Consequently, the primordial tensor spectrum is expected to receive corrections arising from cosmological memory, leading to corresponding modifications of inflationary tensor observables such as the tensor-to-scalar ratio. While the present work introduces a general parametrization of these effects, deriving the precise form of the non-Markovian correction $\delta_{\mathrm{mem}}(k)$ and its scale dependence requires a more fundamental understanding of the microscopic dynamics governing information transfer across the conformal boundary. Developing such a theory, together with detailed predictions for future CMB polarization and gravitational-wave observations, constitutes an important direction for future work.

Beyond its implications for the cosmological constant, the present framework also modifies the interpretation of dynamical dark-energy models in the early Universe. We have argued that consistency of the inherited dark-energy sector excludes a non-phantom asymptotic equation of state for the preceding aeon, since such a branch leads to a divergent inherited dark-energy density at the conformal crossover. In particular, the BBN analysis of EDE [19, 20] carried out in [27] acquires a different physical interpretation. Rather than constraining the EOS of the fundamental quintessence field [21–23] directly, the BBN bounds apply to the effective EOS parameter, which receives contributions from the inherited dark-energy component of the previous aeon, the newly generated quintessence field, and the conformal dark-energy contribution associated with the crossover. Consequently, the phenomenologically preferred interval obtained by [27] constrains the effective combination of parameters appearing in eqn.(37) introducing an intrinsic parameter degeneracy that is absent in conventional EDE scenarios.

This feature provides an important theoretical advantage. In conventional quintessence-based EDE models that alleviate the H_0tension, the scalar field remains Hubble frozen for redshifts $z > z_c$, where the critical redshift $z_c \sim 10^3 - 10^4$ is chosen so that the EDE component becomes dynamically relevant only shortly before recombination. Since the BBN epoch occurs at $z_{\rm BBN} >> z_c$, these models predict $\omega_{\rm quintessnece} \simeq -1$ throughout BBN. Interpreting the BBN constraint of [27] as applying directly to the quintessence field would instead require the field to begin thawing at redshifts many orders of magnitude larger than those conventionally employed in EDE models. Within the present framework, however, no such modification of the quintessence dynamics is required. The effective EDE EOS constrained by BBN receives additional contributions from both the inherited dark-energy component and the conformally induced dynamical dark-energy sector. Consequently, the underlying quintessence field may remain Hubble frozen throughout the BBN epoch, while the effective equation of state simultaneously satisfies the observational constraints. This reconciles the standard EDE mechanism for alleviating the H_0 tension with the recent BBN bounds on the effective EDE EOS.

In summary, the present work should be viewed primarily as an exploratory investigation into the role that dark energy may play within the framework of CCC. Rather than presenting a complete theory, our aim has been to identify several new directions through which the dark sector of CCC may be extended beyond its orthodox formulation. On the one hand, we have argued that the conformal crossover naturally gives rise to an effective cosmological constant in the new aeon, providing a new perspective on the cosmological constant problem and suggesting that non-Markovian correlations between successive aeons may further influence its evolution. On the other hand, we have demonstrated that the effective Early Dark Energy sector in CCC possesses a richer structure than in conventional quintessence models, leading to an effective EOS that need not coincide with that of the underlying scalar field. Consequently, observational constraints from BBN [27] apply to the effective EDE sector rather than to the quintessence field alone. This provides a natural mechanism for reconciling the recent BBN constraints with the conventional EDE picture, allowing the quintessence field to remain Hubble frozen until the critical redshift required to alleviate the H_0 tension while the effective EOS remains compatible with current observations.

Many aspects of this picture remain to be developed. A more fundamental description of the conformal dynamics responsible for the evolution of the effective cosmological constant, a rigorous treatment of inter-aeonic correlations all constitute important avenues for future work. Furthermore, it would be particularly interesting to investigate entropy curves [14, 29] of the universe within the present non-Markovian framework. Since the present framework allows successive aeons to retain a controlled memory of their predecessors, it provides a qualitatively new setting in which the accumulation, transfer, and possible resetting of entropy across conformal crossovers may be studied. Understanding the interplay between cosmological memory, information transfer, and the thermodynamic arrow of time could therefore provide new insights into one of the longstanding conceptual challenges of CCC. More broadly, we hope that the ideas presented here motivate further investigations into the dark-energy sector of CCC.

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A Conserved Quantities in Conformal Cyclic Cosmology

In order to discuss conserved quantities in CCC [15], it is found to be simplest to utilize a two-spinor formalism. Following the convention of [40, 41] , we shall denote the 2-spinor indices by capital italic Latin letters, primed when referring only to the complex-conjugate spin space, and the formal index translation from tensor to 2-spinor form is:

$$a = AA', b = BB', c = CC', d = DD', \ldots \tag{48}$$

and the metric tensor translates to the product of skew-symmetric spinor epsilons according to

$$g_{ab} = \epsilon_{AB}\epsilon_{A'B'}, \quad g^{ab} = \epsilon^{AB}\epsilon^{A'B'}, \tag{49}$$

where

$$\epsilon^{AB} = -\epsilon^{BA}, \quad \epsilon_{AB} = -\epsilon_{BA}, \tag{50}$$

and analogously for primed indices. Moreover, $\epsilon_A^B = -\epsilon_A^B$ and $\epsilon_{A'}^{B'} = -\epsilon_{A'}^{B'} = -\epsilon_{A'}^{B'}$ are the Kronecker delta symbols.

Spinor indices can be raised and lowered according to

$$X^A = \epsilon^{AB}X_B, \quad X_B = X^A\epsilon_{\mathcal{AB}} \tag{51}$$

and analogously for primed indices. Under the conformal rescaling, we take the abstract-indexed epsilon spinors to scale as

$$\epsilon_{AB} \to \tilde{\epsilon}_{AB} = \Omega^2\epsilon_{AB}, \quad \epsilon^{AB} \to \tilde{\epsilon}^{AB} = \Omega^{-2}\epsilon^{AB}, \tag{52}$$

and analogously for primed indices.

In spinor terms, the Weyl tensor takes the form

$$C_{abcd} = \Psi_{ABCD}\epsilon_{A'B'}\epsilon_{C'D'} + \epsilon_{AB}\epsilon_{CD}\bar{\Psi}_{A'B'C'D'}, \tag{53}$$

where the somewhat complicated symmetry and trace-free conditions satisfied by C_{abcd} are now replaced by the total symmetry of Ψ_{ABCD}:

$$\Psi_{ABCD} = \Psi_{(ABCD)}. \tag{54}$$

Moreover, in the vacuum space-time (with cosmological constant Λ),

$$R_{ab} = \Lambda g_{ab}, \tag{55}$$

we find that the Bianchi identity can be expressed as

$$\nabla^{AA'}\Psi_{ABCD} = 0. \tag{56}$$

The above equation is the equation of the wave function of a massless field of spin 2 [42, 43]. The helicty $s = +2$ part of the wavefunction would be described by

$$\nabla^{AA'}\hat{\Psi}_{A'B'C'D'} = 0. \tag{57}$$

For a real spacetime, as opposed to a quantum wave-function, we would have

$$\hat{\Psi}_{ABCD} = \bar{\Psi}_{A'B'C'D'}. \tag{58}$$

However, if when we are indeed dealing with a graviton wave function, we can use complex quantities like Ψ and $\hat{\Psi}$ to describe the negative and positive parts of a graviton's wave function.

We may compare this with the 2-spinor form of Maxwell's equation in free-space,

$$\nabla^{AA'}\psi_{AB} = 0, \tag{59}$$

(with $\psi_{AB} = \psi_{BA}$) where the Maxwell field tensor is

$$F_{ab} = \psi_{AB}\epsilon_{A'B'} + \epsilon_{AB}\bar{\psi}_{A'B'}. \tag{60}$$

Again, we can consider wave-function, this time for photons for which we would need to split eqn.(60) into negative ($s = -1$) and positive ($s = +1$) helicities. For spin $n/2$ (with $n \neq 0$), in conformally flat space-time $\mathcal{M}$, we can consider the equation

$$\nabla^{AA'}\phi_{ABC\ldots F} = 0, \tag{61}$$

where $\phi_{ABC..F}$ has n symmetric indices, and we may take eqn.(61) to be the wave equation for helicity $s = n/2$. Eqn.(61) is conformally invariant provided that $\phi_{ABC\ldots F}$ has a conformal weight -1, where we say that a quantity Q has a conformal weight w if it conformally scales as

$$Q^{\cdots}_{\cdots} \to \Omega^{2w} Q^{\cdots}_{\cdots}. \tag{62}$$

Note that although, in vacuum, the Weyl spinor Ψ_{ABCD} as defined in eqn.(53), satisfies the same eqn.(56) as that shown in (61), it has a conformal weight, as determined by eqn.(53), which is 0, rather than the -1 that would be required for the conformal invariance of eqn.(56). To accommodate this apparent discrepancy, it will be very significant for us to define a quantity ψ_{ABCD} which, in the physical metric g_{ab} of $\mathcal{M}$, is equal to Ψ_{ABCD}:

$$\psi_{ABCD} = \Psi_{ABCD}, \tag{63}$$

but where, after the conformal rescaling to the metric $g^{(n+1)}_{ab} = \Omega^4 g^{(n)}_{ab}$, we demand

$$\tilde{\psi}_{ABCD} = \Omega^{-2}\tilde{\Psi}_{ABCD}. \tag{64}$$

Then ψ_{ABCD} has the required conformal weight -1 needed for the conformal invariance of eqn(61), with ψ_{ABCD} in place of ϕ_{ABCD}.

We shall now introduce the notion of a twistor which, in a conformally flat space-time, is a helicity raising operator, or in the current context a spin-lowering operator. A twistor may be regarded as a solution $\bar{\omega}^A$ of the twistor equation

$$\nabla^{(A}_{A'}\bar{\omega}^{B)} = 0, \tag{65}$$

which is conformally invariant, with $\bar{\omega}^A$ having conformal weight 0. Eqn.(65) has 4 independent solutions in a conformally flat space-time $\mathcal{N}$. The solution space of eqn.(65) is thus a 4-dimensional complex vector space $\mathbb{T}$, referred to as a twistor space of $\mathcal{N}$. Each element $\bar{\omega}^A$ of $\mathbb{T}$ serves to transform any solution $\phi_{ABC\ldots EF}$ of eqn.(61), for spin $n \geq 1$ (i.e with atleast two indices) into another solution

$$\phi_{ABC\ldots EF}\bar{\omega}^F, \tag{66}$$

of the massless free-field equation (61), but with one index fewer.

Let us now apply this procedure twice, in succession to, in particular, a symmetric 4-indexed quantity ψ_{ABCD}, subject to the massles free-feild equation

$$\nabla^{AA'}\psi_{ABCD} = 0, \tag{67}$$

by using two different solutions ω^C and $\hat{\bar{\omega}}^D$ of eqn.(65). We then find

$$\phi_{AB} = \psi_{ABCD}\bar{\omega}^C\hat{\bar{\omega}}^D \tag{68}$$

satisfies the free Maxwell equation

$$\nabla^{AA'}\phi_{AB} = 0, \tag{69}$$

where we have came down from spin 2 to spin 1 in two steps of $\frac{1}{2}$. However, it is more economical to lower the spin by 1 in a single step, by using what would be a symmetric rank 2 twistor, rather than two independent rank 1 twistors. The equation for a symmetric rank-2 twistor (sometimes refereed to as a "Killing spinor" [40, 41, 44]) is

$$\nabla^{(A}_{A'}\kappa^{BC)} = 0 \tag{70}$$

where $\kappa^{AB} = \kappa^{BA}$, which is a conformally invariant equation, κ^{AB} having conformal weight 0. Accordingly, we can replace eqn.(68) by

$$\phi_{AB} = \psi_{ABCD}\kappa^{CD}. \tag{71}$$

and the free Maxwell equation (69) again follows from eqn.(67).

Let us now take a conformally flat space-time $\mathcal{N}$, under consideration here, to be the flat Minkowski 4-space $\mathbb{M}$, where we are concerned with a weak-field gravitational perturbation away from $\mathbb{M}$, described by ψ_{ABCD} (i.e the weak field version of Ψ_{ABCD}), in a vacuum sub-region $\mathcal{V}$ of $\mathbb{M}$, so that eqn.(67) holds true within $\mathcal{V}$. We take $\mathcal{V}$ to surround the gravitational source we are interested in. The idea here is that this source will arise as a source of the free Maxwell field ϕ_{AB} of eqn.(71), for suitable κ^{AB}. The strength of the elctromagnetic source can be calculated by performing an appropriate Gauss integral of ϕ_{AB}, tacken over a closed 2-surface $\mathcal{S}$, lying within this source free region $\mathcal{V}$. This allows us to obtain the total charge, for this field, that is surrounded by $\mathcal{S}$. Such an integral could take the form

$$\int \phi_{AB}\epsilon_{A'B'}dx^a dx^b. \tag{72}$$

Essentially, the integrand is a 2-form, i.e a skew symmetric tensor, say $p_{ab}(=p_{[a,b]})$, which is here given by

$$p_{ab} = \phi_{AB}\epsilon_{A'B'}. \tag{73}$$

If this 2-form is closed, i.e

$$\nabla_{[a}p_{bc]} = 0, \tag{74}$$

then the integral eqn.(72), of p_{ab}, over the closed 2-surface $\mathcal{S}$ remains constant as $\mathcal{S}$ is continuously moved around a region where eqn.(74) remains true, as would necessarily be if the expression (72) is to represent some source quantity external to, but surrounded by, the region $\mathcal{V}$. This is indeed the case here, because eqn.(74) is equivalent to

$$\nabla_a p_{bc}e^{abcd} = 0, \tag{75}$$

where e_{abcd} is the completely skew-symmetrical tensor ($e^{abcd} = e^{[abcd]}$ where $e_{0123} = 1$). In 2-spinor terms, we find

$$e^{abcd} = i\epsilon^{AC}\epsilon^{BD}\epsilon^{A'D'}\epsilon^{B'C'} - i\epsilon^{AD}\epsilon^{BC}\epsilon^{A'C'}\epsilon^{B'D'}, \tag{76}$$

so the expression eqn.(75) indeed vanishes when eqn.(69) holds, because the latter actually asserts that

$$\nabla^{A'}_{A}\phi_{BC} \tag{77}$$

is symmetric in AB and therefore totally symmetric in ABC, so that when contracted with eqn(76) both terms vanish by a skew symmetry of ϵ^{AC} or ϵ^{BC} in eqn.(76). Hence, the integral given by eqn.(72) naturally aquires the interpretation of a "conserved mass" across aeons.

Let us now take x^a to represent the Minkowskian position vector of a variable point under consideration. We shall now define κ^{AB} by

$$\kappa^{(n+1)AB} = x^{(A}_{C'}t^{B)C'}, \tag{78}$$

for which the satisfaction of eqn.(70) immediately follows from

$$\nabla^{(n+1)}_{a}x^b = \delta^b_a, \quad \text{i.e.}\nabla^{(n+1)}_{AA'}x^{BB'} = \epsilon^{(n+1)B}_{A}\epsilon^{(n+1)B'}_{A'}. \tag{79}$$

The conserved mass across aeons now reads

$$M = \int \phi_{AB}\epsilon_{A'B'}dx^a dx^b = \int \psi_{ABCD}\kappa^{CD}\epsilon_{A'B'}dx^a dx^b = \int \psi_{ABCD}x^{(A}_{E'}t^{B)E'}\epsilon_{A'B'}dx^a dx^b. \tag{80}$$

Now, for a Schwarzchild solution of mass m, we have

$$\tilde{\psi}_{ABCD} = -\frac{6mG}{r^3}o_{(A}o_B i_C i_{D)}, \tag{81}$$

where G is the Gravitational Constant and where, at each point of $\mathbb{M}$ we introduce a spin frame (o_A, i_A), with $o_A i^A = -1$, where the flagpole of o_A points in the outward null direction and the flagpole of i_A in the inward null direction. As far as the choice of flagpoles, i.e plases, of these spinors, this is unimportant since in all the expression of relevance these spinors appear in pairs where phases cancel out.

In accordance with standard procedures relating spin frames with Minkowskian orthonormal frames, we can arrange the scalings of o^A and i^A so that

$$t^a = \frac{1}{\sqrt{2}}(a^A o^{A'} + i^A i^{A'}), \tag{82}$$

and at the point with position vector x^a,

$$x^a = \frac{r}{\sqrt{2}}(o^A o^{A'} - i^A i^{A'}) + \tau t^\alpha, \tag{83}$$

where r is the spatial $\mathbb{M}$-distance and τ is the conformal time parameter. Putting these two together, we obtain,

$$\phi^{(n+1)}_{AB} = -\frac{6mG}{r^2}o_{(A}o_B i_C i_{D)}o^C i^D = \frac{mG}{r^2}o_{(A}i_{B)}. \tag{84}$$

Taking $\mathcal{S}$ to be the sphere of radius r in the Euclidean 3-space $\tau =$ cont., we find that

$$\int \phi_{AB}\epsilon_{A'B'}dx^a dx^b = 4\pi mG. \tag{85}$$

Hence, the conserved quantity M across aeons is

$$M = 4\pi mG, \tag{86}$$

since $4\pi G$ is just a constant, eqn.(86) implies that implies that the energy content of the aeon remains invariant after conformal rescaling and is "carried through" in the next aeon, as claimed in Section 2.

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